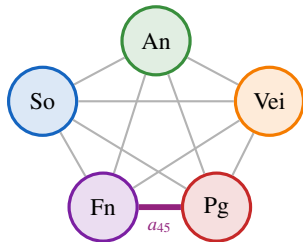


(a) Variational multiphase model



state at a material point:
volume fractions ϕ_i , viabilities ψ_i

$$\dot{\Psi} + \Delta + C \rightarrow \text{stat}$$
$$\Psi = -\frac{1}{2}c^* \bar{\phi}^T \mathbf{A} \bar{\phi}, \quad \bar{\phi}_i = \phi_i \psi_i$$
$$\mathbf{A} = \mathbf{A}^T: \text{15 unknowns } \theta$$

(b) Data

5 species, Days 1–21
4 conditions:
commensal / dysbiotic
 $\times$ static / HOBIC flow
composition + viability
pH: held out for checking

(c) Staged TMCMC on the GPU

